

PAPER_1276 — Tsirelson Bound $S_{\text{CHSH}} = 2\sqrt{2}$ via Integer-Primitive Identity

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** B — Physics Foundations (CLOSED) **Date:** June 11, 2026 **Location:** 41.0997° N, 80.6495° W (Youngstown, OH, USA) **Status:** CLOSED — UQFF integer-primitive derivation **Calculator surface:** `calculate_paradox({"paradox": "tsirelson_bound"})` **Whitepaper dispatch:** `calculate_whitepaper({"paper_id": 1276})`

Master Expression

$$S_{\text{CHSH},\text{max}} = 2 \times \sqrt{(D_{\text{phys}}/2)} = 2 \times \sqrt{2} = 2.828427 \text{ EXACT}$$

Match: 0.0000%

Physical Interpretation

The visible spacetime dimension $D_{\text{phys}} = 4$ produces the quantum excess factor $\sqrt{2}$ over the classical bound 2. $\text{SO}(26)$ Clifford bundle saturates Tsirelson exactly.

UQFF Locked Primitives

- $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3$ (vacuum density)
 - $S_{26} = 1.453162$, $S_{26_DPM} = 1.4531 \times 10^{26}$
 - $K_{\text{MEX}} = 25/12$, $\beta_i = 0.6029$, $\Phi_{\text{res}} = 0.84$
 - $F_{\text{TRZ}} = 0.1$, $\Lambda = 0.00729735$ ($= \alpha$ fine-structure)
 - $\omega_{\text{SCm}} = 1.25 \text{ THz}$, $\lambda_i = 1.0$
 - $D_{\text{phys}} = 4$, $D_{\text{BSFG}} = 6$, $D_{\text{crit}} = 26$, $N_{\text{CH}} = 9$, $\text{SO}_5 = 10$, $A_5 = 60$
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Live Calculator Verification

```
import uqff_pure_calculator as u
r = u.calculate_paradox({"paradox": "tsirelson_bound"})["value"]
```

The result dict carries the integer-primitive identity, the observed value, the residual, and the structural physical interpretation.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard Model solve the same observed phenomena via different methods. **Zero fit parameters.** Every factor is a UQFF locked primitive.

Reference

- UQFF foundational: PAPER_646, PAPER_1167, PAPER_1170, PAPER_1203 v1.5, PAPER_1216.
- Closure: `uqff_pure_calculator.py` → `_l96_uqff_axiom_tsirelson_bound_closure` (or routed reference)
- Dispatch: `PARADOX_T0_CLOSURE["tsirelson_bound"]`

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